

Annual Review of Economics

Econometric Methods for Program Evaluation

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Annu. Rev. Econ. 2018. 10:465–503

The *Annual Review of Economics* is online at economics.annualreviews.org

<https://doi.org/10.1146/annurev-economics-080217-053402>

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JEL codes: C1, C2, C3, C54

Keywords

causal inference, policy evaluation, treatment effects

Abstract

Program evaluation methods are widely applied in economics to assess the effects of policy interventions and other treatments of interest. In this article, we describe the main methodological frameworks of the econometrics of program evaluation. In the process, we delineate some of the directions along which this literature is expanding, discuss recent developments, and highlight specific areas where new research may be particularly fruitful.

1. INTRODUCTION

The nature of empirical research in economics has profoundly changed since the emergence in the 1990s of a new way to understand identification in econometric models. This approach emphasizes the importance of heterogeneity across units in the parameters of interest and of the choice of the sources of variation in the data that are used to estimate those parameters. Initial impetus came from empirical research by Angrist, Ashenfelter, Card, Krueger and others, while Angrist, Heckman, Imbens, Manski, Rubin, and others provided many of the early methodological innovations (see, in particular, Angrist 1990, Angrist & Krueger 1991, Angrist et al. 1996, Card 1990, Card & Krueger 1994, Heckman 1990, Manski 1990; for earlier contributions, see Ashenfelter 1978, Ashenfelter & Card 1985, Heckman & Robb 1985).

While this greater emphasis on identification has permeated all brands of econometrics and quantitative social science, nowhere have the effects been felt more profoundly than in the field of program evaluation—a domain expanding the social, biomedical, and behavioral sciences that studies the effects of policy interventions. The policies of interest are often governmental programs, like active labor market interventions or antipoverty programs. In other instances, the term policy is more generally understood to include any intervention of interest by public or private agents or by nature.

In this article, we offer an overview of classical tools and recent developments in the econometrics of program evaluation and discuss potential avenues for future research. Our focus is on ex post evaluation exercises, where the effects of a policy intervention are evaluated after some form of the intervention (such as a regular implementation of the policy, an experimental study, or a pilot study) is deployed and the relevant outcomes under the intervention are measured. Many studies in economics and the social sciences aim to evaluate existing or experimentally deployed interventions, and the methods surveyed in this article have been shown to have broad applicability in these settings. In many other cases, however, economists and social scientists are interested in ex ante evaluation exercises, that is, in estimating the effect of a policy before such policy is implemented on the population of interest (e.g., Todd & Wolpin 2010). In the absence of measures of the outcomes under the intervention of interest, ex ante evaluations aim to extrapolate outside the support of the data. Often, economic models that precisely describe the behavior of economic agents serve as useful extrapolation devices for ex ante evaluations. Methods that are suitable for ex ante evaluation have been previously covered in the *Annual Review of Economics* (Arcidiacono & Ellickson 2011, Berry & Haile 2016).

This distinction between ex ante and ex post program evaluations is closely related to often-discussed differences between the so-called structural and causal inference schools in econometrics. The use of economic models as devices to extrapolate outside the support of the data in ex ante program evaluation is typically associated with the structural school, while ex post evaluations that do not explicitly extrapolate using economic models of behavior are often termed causal or reduced form.¹ Given that they serve related but quite distinct purposes, we find the perceived conflict between these two schools to be rather artificial, and the debates about the purported superiority of one of the approaches over the other largely unproductive.

Our goal in this article is to provide a summary overview of the literature on the econometrics of program evaluation for ex post analysis and, in the process, to delineate some of the directions

¹As argued by Imbens (2010), however, reduced form is a misnomer relative to the literature on simultaneous equation models, where the terms structural form and reduced form have precise meanings. Moreover, terms like structural model or causal inference are nomenclature, rather than exclusive attributes of the literatures and methods that they refer to.

along which it is expanding, discussing recent developments and the areas where research may be particularly fruitful in the near future.

Other recent surveys on the estimation of causal treatment effects and the econometrics of program evaluation from different perspectives and disciplines include those by Abadie (2005a), Angrist & Pischke (2008, 2014), Athey & Imbens (2017c), Blundell & Costa Dias (2009), DiNardo & Lee (2011), Heckman & Vytlacil (2007), Hernán & Robins (2018), Imbens & Rubin (2015), Imbens & Wooldridge (2009), Lee (2016), Manski (2008), Pearl (2009), Rosenbaum (2002, 2010), Van der Laan & Robins (2003), and VanderWeele (2015), among many others.

2. CAUSAL INFERENCE AND PROGRAM EVALUATION

Program evaluation is concerned with the estimation of the causal effects of policy interventions. These policy interventions can be of very different natures depending on the context of the investigation, and they are often generically referred to as treatments. Examples include conditional transfer programs (Behrman et al. 2011), health care interventions (Finkelstein et al. 2012, Newhouse 1996), and large-scale online A/B studies in which IP addresses visiting a particular web page are randomly assigned to different designs or contents (see, e.g., Bakshy et al. 2014).

2.1. Causality and Potential Outcomes

We represent the value of the treatment by the random variable W . We aim to learn the effect of changes in treatment status on some observed outcome variable, denoted by Y . Following Neyman (1923), Rubin (1974), and many others, we use potential outcomes to define causal parameters: Y_w represents the potential value of the outcome when the value of the treatment variable, W , is set to w . For each value of w in the support of W , the potential outcome Y_w is a random variable with a distribution over the population. The realized outcome, Y , is such that, if the value of the treatment is equal to w for a unit in the population, then for that unit, $Y = Y_w$, while other potential outcomes $Y_{w'}$ with $w' \neq w$ remain counterfactual.

A strong assumption lurks implicit in the last statement. Namely, the realized outcome for each particular unit depends only on the value of the treatment of that unit and not on the treatment or on outcome values of other units. This assumption is often referred to as the stable unit treatment value assumption (SUTVA) and rules out interference between units (Rubin 1980). SUTVA is a strong assumption in many practical settings; for example, it may be violated in an educational setting with peer effects. However, concerns about interference between units can often be mitigated through careful study design (see, e.g., Imbens & Rubin 2015).

The concepts of potential and realized outcomes are deeply ingrained in economics. A demand function, for example, represents the potential quantity demanded as a function of price. Quantity demanded is realized for the market price and is counterfactual for other prices.

While, in practice, researchers may be interested in a multiplicity of treatments and outcomes, we abstract from that in our notation, where Y and W are scalar random variables. In addition to treatments and potential outcomes, the population is characterized by covariates X , a $(k \times 1)$ vector of variables that are predetermined relative to the treatment. That is, while X and W may not be independent (perhaps because X causes W , or perhaps because they share common causes), the value of X cannot be changed by active manipulation of W . Often, X contains characteristics of the units measured before W is known.

Although the notation allows the treatment to take on an arbitrary number of values, we introduce additional concepts and notation within the context of a binary treatment, that is, $W \in \{0, 1\}$. In this case, $W = 1$ often denotes exposure to an active intervention (e.g., participation

in an antipoverty program), while $W = 0$ denotes remaining at the status quo. For simplicity of exposition, our discussion mostly focuses on the binary treatment case. The causal effect of the treatment (or treatment effect) can be represented by the difference in potential outcomes, $Y_1 - Y_0$. Potential outcomes and the value of the treatment determine the observed outcome

$$Y = WY_1 + (1 - W)Y_0. \quad 1.$$

Equation 1 represents what is often termed the fundamental problem of causal inference (Holland 1986). The realized outcome, Y , reveals Y_1 if $W = 1$ and Y_0 if $W = 0$. However, the unit-level treatment effect, $Y_1 - Y_0$, depends on both quantities. As a result, the value of $Y_1 - Y_0$ is unidentified from observing (Y, W, X) .

Beyond the individual treatment effects, $Y_1 - Y_0$, which are identified only under assumptions that are not plausible in most empirical settings, the objects of interest, or estimands, in program evaluation are characteristics of the joint distribution of (Y_1, Y_0, W, X) in the sample or in the population. For most of this review, we focus on estimands defined for a certain population of interest from which we have extracted a random sample. Like in the work of Abadie et al. (2017a), we say that an estimand is causal when it depends on the distribution of the potential outcomes, (Y_1, Y_0) beyond its dependence on the distribution of (Y, W, X) . This is in contrast to descriptive estimands, which are characteristics of the distribution of (Y, W, X) .²

The average treatment effect (ATE),

$$\tau_{\text{ATE}} = E[Y_1 - Y_0],$$

and the average treatment effect on the treated (ATET),

$$\tau_{\text{ATET}} = E[Y_1 - Y_0 | W = 1],$$

are causal estimands that are often of interest in the program evaluation literature. Under SUTVA, ATE represents the difference in average outcomes induced by shifting the entire population from the inactive to the active treatment. ATET represents the same object for the treated. To improve the targeting of a program, researchers often aim to estimate ATE or ATET after accounting for a set of units' characteristics, X . Notice that ATE and ATET depend on the distribution of (Y_1, Y_0) beyond their dependence on the distribution of (Y, W, X) . Therefore, they are causal estimands under the definition of Abadie et al. (2017a).

2.2. Confounding

Consider now the descriptive estimand,

$$\tau = E[Y | W = 1] - E[Y | W = 0],$$

which is the difference in average outcomes for the two different treatment values. ATE and ATET are measures of causation, while the difference in means, τ , is a measure of association. Notice that

$$\begin{aligned} \tau &= \tau_{\text{ATE}} + b_{\text{ATE}} \\ &= \tau_{\text{ATET}} + b_{\text{ATET}}, \end{aligned} \quad 2.$$

² Abadie et al. (2017a) provide a discussion of descriptive and causal estimands in the context of regression analysis.

where b_{ATE} and b_{ATET} are bias terms given by

$$b_{ATE} = (E[Y_1|W = 1] - E[Y_1|W = 0]) \Pr(W = 0) \quad 3.$$

$$+ (E[Y_0|W = 1] - E[Y_0|W = 0]) \Pr(W = 1) \quad 4.$$

and

$$b_{ATET} = E[Y_0|W = 1] - E[Y_0|W = 0]. \quad 5.$$

Equations 2–5 combine into a mathematical representation of the often-invoked statement that association does not imply causation, enriched with the statement that lack of association does not imply lack of causation. If average potential outcomes are identical for treated and nontreated units, an untestable condition, then the bias terms b_{ATE} and b_{ATET} disappear. However, if potential outcomes are not independent of the treatment, then, in general, the difference in mean outcomes between the treated and the untreated is not equal to the ATE or ATET. Lack of independence between the treatment and the potential outcomes is often referred to as confounding. Confounding may arise, for example, when information that is correlated with potential outcomes is used for treatment assignment or when agents actively self-select into treatment based on their potential gains.

Confounding is a powerful notion, as it explains departures of descriptive estimands from causal ones. Directed acyclic graphs (DAGs; see Pearl 2009) provide graphical representations of causal relationships and confounding.

A DAG is a collection of nodes and directed edges among nodes, with no directed cycles. Nodes represent random variables, while directed edges represent causal effects not mediated by other variables in the DAG. Moreover, a causal DAG must contain all causal effects among the variables in the DAG, as well as all variables that are common causes of any pair of variables in the DAG (even if common causes are unobserved).

Consider the DAG in **Figure 1**. This DAG includes a treatment, W , and an outcome of interest, Y . In this DAG, W is not a cause of Y , as there is no directed edge from W to Y . The DAG also includes a confounder, U . This confounder is a common cause of W and Y . Even when W does not cause Y , these two variables may be correlated because of the confounding effect of U . A familiar example of this structure in economics is Spence's (1973) job-market signaling model, where W is worker education; Y is worker productivity; and there are other worker characteristics, U , that cause education and productivity (e.g., worker attributes that increase productivity and reduce the cost of acquiring education). In this example, even when education does not cause worker productivity, the two variables may be positively correlated.

The DAG in **Figure 1** implies that Y_w does not vary with w because W does not cause Y . However, Y and W are not independent, as they share a common cause, U . Confounding may arise in more complicated scenarios. In Section 4, we cover other forms of confounding that may arise in DAGs. Confounding, of course, often coexists with true causal effects. That is, the DAG in **Figure 1** implies the presence of confounding regardless of whether there is a directed edge from W to Y .



Figure 1

Confounding in a directed acyclic graph. The graph includes a treatment, W ; an outcome of interest, Y ; and a confounder, U .

Beyond average treatment effects, other treatment parameters of interest depend on the distribution of (Y_1, Y_0) . These include treatment effects on characteristics of the distribution of the outcome other than the average (e.g., quantiles), as well as characteristics of the distribution of the treatment effect, $Y_1 - Y_0$ (e.g., variance), other than its average. In Section 3, we elaborate further on the distinction between treatment effects on the distribution of the outcome versus characteristics of the distribution of the treatment effect.

3. RANDOMIZED EXPERIMENTS

Active randomization of the treatment provides the basis of what is arguably the most successful and extended scientific research design for program evaluation. Often referred to as the gold standard of scientific evidence, randomized clinical trials play a central role in the natural sciences, as well as in the drug approval process in the United States, Europe, and elsewhere (Bothwell et al. 2016). While randomized assignment is often viewed as the basis of a high standard of scientific evidence, this view is by no means universal. Cartwright (2007), Heckman & Vytlačil (2007), and Deaton (2010), among others, challenge the notion that randomized experiments occupy a preeminent position in the hierarchy of scientific evidence, while Angrist & Pischke (2010), Imbens (2010), and others emphasize the advantages of experimental designs.

3.1. Identification Through Randomized Assignment

Randomization assigns treatment using the outcome of a procedure (natural, mechanical, or electronic) that is unrelated to the characteristics of the units and, in particular, unrelated to potential outcomes. Whether randomization is based on coin tossing, random number generation in a computer, or the radioactive decay of materials, the only requirement of randomization is that the generated treatment variable is statistically independent of potential outcomes. Complete (or unconditional) randomization implies

$$(Y_1, Y_0) \perp\!\!\!\perp W, \quad 6.$$

where the symbol $\perp\!\!\!\perp$ denotes statistical independence. If Equation 6 holds, we say that the treatment assignment is unconfounded. In Section 4, we consider conditional versions of unconfoundedness where the value of the treatment W for each particular unit may depend on certain characteristics of the units, and Equation 6 holds conditional on those characteristics.

The immediate consequence of randomization is that the bias terms in Equations 4 and 5 are equal to zero:

$$b_{ATE} = b_{ATET} = 0,$$

which, in turn, implies

$$\tau_{ATE} = \tau_{ATET} = \tau = E[Y|W = 1] - E[Y|W = 0]. \quad 7.$$

Beyond average treatment effects, randomization identifies any characteristic of the marginal distributions of Y_1 and Y_0 . In particular, the cumulative distributions of potential outcomes $F_{Y_1}(y) = \Pr(Y_1 \leq y)$ and $F_{Y_0}(y) = \Pr(Y_0 \leq y)$ are identified by

$$F_{Y_1}(y) = \Pr(Y \leq y|W = 1)$$

and

$$F_{Y_0}(y) = \Pr(Y \leq y|W = 0).$$

While the average effect of a treatment can be easily described using τ_{ATE} , it is more complicated to establish a comparison between the entire distributions of Y_1 and Y_0 . To be concrete, suppose that the outcome variable of interest, Y , is income. Then, an analyst may want to rank the income distributions under the active intervention and in the absence of the active intervention. For that purpose, it is useful to consider the following distributional relationships: (a) equality of distributions, $F_{Y_1}(y) = F_{Y_0}(y)$ for all $y \geq 0$; (b) first-order stochastic dominance (with F_{Y_1} dominating F_{Y_0}), $F_{Y_1}(y) - F_{Y_0}(y) \leq 0$ for all $y \geq 0$; and (c) second-order stochastic dominance (with F_{Y_1} dominating F_{Y_0}), $\int_0^y (F_{Y_1}(z) - F_{Y_0}(z)) dz \leq 0$ for all $y \geq 0$.

Equality of distributions implies that the treatment has no effect on the distribution of the outcome variable, in this case income. It implies $\tau_{ATE} = 0$ but represents a stronger notion of null effect. Under mild assumptions, first- and second-order stochastic dominance imply that the income distribution under the active treatment is preferred to the distribution of income in the absence of the active treatment (see, e.g., Abadie 2002, Foster & Shorrocks 1988).

Equivalently, the distributions of Y_1 and Y_0 can be described by their quantiles. Quantiles of Y_1 and Y_0 are identified by inverting F_{Y_1} and F_{Y_0} or, more directly, by

$$\begin{aligned} Q_{Y_1}(\theta) &= Q_{Y|W=1}(\theta), \\ Q_{Y_0}(\theta) &= Q_{Y|W=0}(\theta), \end{aligned}$$

with $\theta \in (0, 1)$, where $Q_Y(\theta)$ denotes the θ quantile of the distribution of Y . The effect of the treatment on the θ quantile of the outcome distribution is given by the quantile treatment effect,

$$Q_{Y_1}(\theta) - Q_{Y_0}(\theta). \quad 8.$$

Notice that, while quantile treatment effects are identified from Equation 6, quantiles of the distribution of the treatment effect, $Q_{Y_1-Y_0}(\theta)$, are not identified. Unlike expectations, quantiles are nonlinear operators, so in general, $Q_{Y_1-Y_0}(\theta) \neq Q_{Y_1}(\theta) - Q_{Y_0}(\theta)$. Moreover, while quantile treatment effects depend only on the marginal distributions of Y_1 and Y_0 , which are identified from Equation 6, quantiles of the distribution of treatment effects are functionals of the joint distribution of (Y_1, Y_0) , which involves information beyond the marginals. Because, even in a randomized experiment, the two potential outcomes are never both realized for the same unit, the joint distribution of (Y_1, Y_0) is not identified. As a result, quantiles of $Y_1 - Y_0$ are not identified even in the context of a randomized experiment.

To gain intuitive understanding of this lack of identification result, consider an example where the marginal distributions of Y_1 and Y_0 are identical, symmetric around zero, and nondegenerate. Identical marginals are consistent with a null treatment effect, that is, $Y_1 - Y_0 = 0$ with probability one. In this scenario, all treatment effects are zero. However, identical marginals and symmetry around zero are also consistent with $Y_1 = -Y_0$ or, equivalently, $Y_1 - Y_0 = -2Y_0$ with probability one, which leads to positive treatment effects for half of the population and negative treatment effects for the other half. In the first scenario, the treatment does not change the value of the outcome for any unit in the population. In contrast, in the second scenario, the locations of units within the distribution of the outcome variable are reshuffled, but the shape of the distribution does not change. While these two different scenarios imply different distributions of the treatment effect, $Y_1 - Y_0$, they are both consistent with the same marginal distributions of the potential outcomes, Y_1 and Y_0 , so the distribution of $Y_1 - Y_0$ is not identified.

Although the distribution of treatment effects is not identified from randomized assignment of the treatment, knowledge of the marginal distributions of Y_1 and Y_0 can be used to bound functionals of the distribution of $Y_1 - Y_0$ (see, e.g., Fan & Park 2010, Firpo & Ridder 2008). In

particular, if the distribution of Y is continuous, then for each $t \in \mathbb{R}$,

$$\max \left\{ \sup_{y \in \mathbb{R}} [F_{Y_1}(y) - F_{Y_0}(y - t)], 0 \right\} \leq F_{Y_1 - Y_0}(t) \leq 1 + \min \left\{ \inf_{y \in \mathbb{R}} [F_{Y_1}(y) - F_{Y_0}(y - t)], 0 \right\} \quad 9.$$

provides sharp bounds on the distribution of $Y_1 - Y_0$. This expression can be applied to bound the fraction of units with positive (or negative) treatment effects. Bounds on $Q_{Y_1 - Y_0}(\theta)$ can be obtained by inversion.

3.2. Estimation and Inference in Randomized Studies

So far, we have concentrated on identification issues. We now turn to estimation and inference. Multiple modes of statistical inference are available for treatment effects (see Rubin 1990), and our discussion mainly focuses on two of them: (a) sampling-based frequentist inference and (b) randomization-based inference on a sharp null. We also briefly mention permutation-based inference. However, because of space limitations, we omit Bayesian methods (see Rubin 1978, 2005) and finite-sample frequentist methods based on randomization (see Abadie et al. 2017a, Neyman 1923). Imbens & Rubin (2015) give an overview of many of these methods.

3.2.1. Sampling-based frequentist inference. In the context of sampling-based frequentist inference, we assume that data consist of a random sample of treated and nontreated units, and that the treatment has been assigned randomly. For each unit i in the sample, we observe the value of the outcome, Y_i ; the value of the treatment, W_i ; and, possibly, the values of a set of covariates, X_i . Consider the common scenario where randomization is carried out with a fixed number of treated units, n_1 , and untreated units, n_0 , such that $n = n_1 + n_0$ is the total sample size. We construct estimators using the analogy principle (Manski 1988). That is, estimators are sample counterparts of population objects that identify parameters of interest. In particular, Equation 7 motivates estimating $\tau = \tau_{ATE} = \tau_{ATET}$ using the difference in sample means of the outcome between treated and nontreated,

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n W_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - W_i) Y_i.$$

Under conventional regularity conditions, a large sample approximation to the sampling distribution of $\hat{\tau}$ is given by

$$\frac{\hat{\tau} - \tau}{\widehat{\text{se}}(\hat{\tau})} \xrightarrow{d} \mathcal{N}(0, 1), \quad 10.$$

where

$$\widehat{\text{se}}(\hat{\tau}) = \left(\frac{\hat{\sigma}_{Y|W=1}^2}{n_1} + \frac{\hat{\sigma}_{Y|W=0}^2}{n_0} \right)^{1/2}, \quad 11.$$

and $\hat{\sigma}_{Y|W=1}^2$ and $\hat{\sigma}_{Y|W=0}^2$ are the sample counterparts of the conditional variance of Y given $W = 1$ and $W = 0$, respectively.

The estimator $\hat{\tau}$ coincides with the coefficient on W from a regression of Y on W and a constant, and $\widehat{\text{se}}(\hat{\tau})$ is the corresponding heteroskedasticity-robust standard error. Inference based on Equations 10 and 11 has an asymptotic justification but can perform poorly in finite samples, especially if n_1 and n_0 differ considerably. Imbens & Kolesár (2016) discuss small sample adjustments to the t -statistic and its distribution and demonstrate that such adjustments substantially improve coverage rates of confidence intervals in finite samples. In addition, more general sampling processes (e.g., clustered sampling) or randomization schemes (e.g., stratified randomization) are

possible. They affect the variability of the estimator and, therefore, the standard error formulas (for details, see Abadie et al. 2017b, Imbens & Rubin 2015).

Estimators of other quantities of interest discussed in Section 2 can be similarly constructed using sample analogs. In particular, for any $\theta \in (0, 1)$, the quantile treatment effects estimator is given by

$$\widehat{Q}_{Y|W=1}(\theta) - \widehat{Q}_{Y|W=0}(\theta),$$

where $\widehat{Q}_{Y|W=1}(\theta)$ and $\widehat{Q}_{Y|W=0}(\theta)$ are the sample analogs of $Q_{Y|W=1}(\theta)$ and $Q_{Y|W=0}(\theta)$, respectively. Sampling-based frequentist inference on quantile treatment effects follows from the results of Koenker & Bassett (1978) on quantile inference for quantile regression estimators. Bounds on the distribution of $F_{Y_1 - Y_0}$ can be computed by replacing the cumulative functions $F_{Y|W=1}$ and $F_{Y|W=0}$ in Equation 9 with their sample analogs, $\widehat{F}_{Y|W=1}$ and $\widehat{F}_{Y|W=0}$, respectively. The estimators of the cumulative distribution functions F_{Y_1} and F_{Y_0} (or, equivalently, estimators of the quantile functions Q_{Y_1} and Q_{Y_0}) can also be applied to test for equality of distributions and first- and second-order stochastic dominance (see, e.g., Abadie 2002).

3.2.2. Randomization inference on a sharp null. The testing exercise in Section 3.2.1 is based on sampling inference. That is, it is based on the comparison of the value of a test statistic for the sample at hand to its sampling distribution under the null hypothesis. In contrast, randomization inference takes the sample as fixed and concentrates on the null distribution of the test statistic induced by the randomization of the treatment. Randomization inference originated with Fisher's (1935) proposal to use the physical act of randomization as a reasoned basis for inference.

Consider the example in Panel A of **Table 1**. This panel shows the result of a hypothetical experiment where a researcher randomly selects four individuals out of a sample of eight individuals to receive the treatment and excludes the remaining four individuals from the treatment. For each of the eight sample individuals, $i = 1, \dots, 8$, the researcher observes treatment status, W_i , and the value of the outcome variable, Y_i . For concreteness, we assume that the researcher adopts the difference in mean outcomes between the treated and the nontreated as a test statistic for randomization inference. The value of this statistic in the experiment is $\widehat{\tau} = 6$.

The sample observed by the researcher is informative only about the value of $\widehat{\tau}$ obtained under one particular realization of the randomized treatment assignment: the one observed in practice. There are, however, 70 possible ways to assign four individuals out of eight to the treatment group. Let Ω be the set of all possible randomization realizations. Each element ω of Ω has probability $1/70$. Consider Fisher's null hypothesis that the treatment does not have any effect on the outcomes of any unit, that is, $Y_{1i} = Y_{0i}$ for each experimental unit. Notice that Fisher's null pertains to the

Table 1 Randomization distribution of a difference in means

Panel A: Sample and sample statistic									
Y_i	12	4	6	10	6	0	1	1	
W_i	1	1	1	1	0	0	0	0	$\widehat{\tau} = 6$
Panel B: Randomization distribution									$\widehat{\tau}(\omega)$
$\omega = 1$	1	1	1	1	0	0	0	0	6
$\omega = 2$	1	1	1	0	1	0	0	0	4
$\omega = 3$	1	1	1	0	0	1	0	0	1
$\omega = 4$	1	1	1	0	0	0	1	0	1.5
$\dots \omega = 70$	0	0	0	0	1	1	1	1	-6

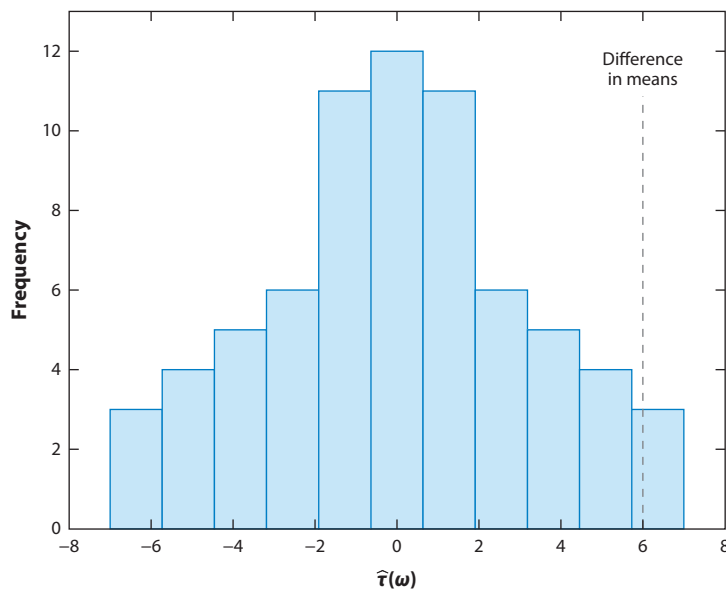


Figure 2

Randomization distribution of the difference in means. The vertical line represents the sample value of $\hat{\tau}$.

sample and not necessarily to the population. Under Fisher's null, it is possible to calculate the value $\hat{\tau}(\omega)$ that would have been observed for each possible realization of the randomized assignment, $\omega \in \Omega$. The distribution of $\hat{\tau}(\omega)$ in Panel B of **Table 1** is called the randomization distribution of $\hat{\tau}$. A histogram of the randomization distribution of $\hat{\tau}$ is depicted in **Figure 2**. p -values for one-sided and two-sided tests are calculated in the usual fashion: $\Pr(\hat{\tau}(\omega) \geq \hat{\tau}) = 0.0429$ and $\Pr(|\hat{\tau}(\omega)| \geq |\hat{\tau}|) = 0.0857$, respectively. These probabilities are calculated over the randomization distribution. Because the randomization distribution under the null can be computed without error, these p -values are exact.³

Fisher's null hypothesis of $Y_{1i} - Y_{0i} = 0$ for every experimental unit is an instance of a sharp null, that is, a null hypothesis under which the values of the treatment effect for each unit in the experimental sample are fixed. Notice that this sharp null, if extended to every population unit, implies but is not implied by Neyman's null, $E[Y_1] - E[Y_0] = 0$. However, a rejection of Neyman's null, using the t -statistic discussed above in the context of sampling-based frequentist inference, does not imply that Fisher's test will reject. This is explained by Ding (2017) using the power properties of the two types of tests and illustrated by Young (2017).

3.2.3. Permutation methods. Some permutation methods are related to randomization tests but rely on a sampling interpretation. They can be employed to evaluate the null hypothesis that Y_1 and Y_0 have the same distribution. Notice that this is not a sharp hypothesis and that it pertains to the population rather than to the sample at hand. These tests are based on the observation that, under the null hypothesis, the distribution of the vector of observed outcomes is invariant

³In a setting with many experimental units, it may be computationally costly to calculate the value of the test statistic under all possible assignments. In those cases, the randomization distribution can be approximated from a randomly selected subset of assignments.

under permutations. Ernst (2004) provides more discussion and comparisons between Fisher's randomization inference and permutation methods, and Bugni et al. (2018), who also include further references on this topic, describe a recent application of permutation-based inference in the context of randomized experiments.

3.3. Recent Developments

Looking ahead, the literature on the analysis and interpretation of randomized experiments remains active, with several interesting developments and challenges still present beyond what is discussed in this section. Athey & Imbens (2017a) provide a detailed introduction to the econometrics of randomized experiments. Some recent topics in this literature include (a) decision theoretic approaches to randomized experiments and related settings (Banerjee et al. 2017); (b) design of complex or high-dimensional experiments accounting for new (big) data environments such as social media or networks settings (Bakshy et al. 2014); (c) the role of multiple hypothesis testing and alternative inference methods, such as model selection, shrinkage, and empirical Bayes approaches (see, e.g., Abadie & Kasy 2017); and (d) subgroup, dynamic, and optimal treatment effect analysis, as well as related issues of endogenous stratification (see, e.g., Abadie et al. 2017c, Athey & Imbens 2017b, Murphy 2003).

4. CONDITIONING ON OBSERVABLES

4.1. Identification by Conditional Independence

In the absence of randomization, the independence condition in Equation 6 is rarely justified, and the estimation of treatment effect parameters requires additional identifying assumptions. In this section, we discuss methods based on a conditional version of the unconfoundedness assumption in Equation 6. We assume that potential outcomes are independent of the treatment after conditioning on a set of observed covariates. The key underlying idea is that confounding, if present, is fully accounted for by observed covariates.

Consider an example where newly admitted college students are awarded a scholarship on the basis of the result of a college entry exam, X , and other student characteristics, V . We use the binary variable W to code the receipt of a scholarship award. To investigate the effect of the award on college grades, Y , one may be concerned about the potential confounding effect of precollege academic ability, U , which may not be precisely measured by X . Confounding may also arise if V has a direct effect on Y .

This setting is represented in the DAG in **Figure 3a**. Precollege academic ability, U , is a cause of W , the scholarship award, through its effect on the college entry exam grade, X . Precollege ability, U , is also a direct cause of college academic grades, Y . The confounding effect of U induces statistical dependence between W and Y that is not reflective of the causal effect of W on Y . In a DAG, a path is a collection of consecutive edges (e.g., $X \rightarrow W \rightarrow Y$), and confounding can generate from backdoor paths, that is, paths from the treatment, W , to the outcome, Y , that start with incoming arrows. The path $W \leftarrow X \leftarrow U \rightarrow Y$ is a backdoor path in **Figure 3a**. An additional backdoor path, $W \leftarrow V \rightarrow Y$, emerges if other causes of the award, aside from the college entry exam, also have a direct causal effect on college grades. The presence of backdoor paths may create confounding, invalidating Equation 6.

Consider a conditional version of unconfoundedness,

$$(Y_1, Y_0) \perp\!\!\!\perp W | X. \quad 12.$$

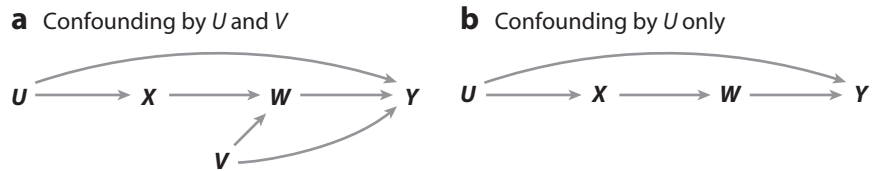


Figure 3

Causal effect and confounding. The graphs show the result of a college entrance exam, X ; other student characteristics affecting college admission, V ; receipt of a scholarship award, W ; precollege academic ability, U ; and college grades, Y . (a) Confounding by U and V . (b) Confounding by U alone.

Equation 12 states that (Y_1, Y_0) is independent of W given X .⁴ This assumption allows identification of treatment effect parameters by conditioning on X . That is, controlling for X makes the treatment unconfounded.⁵ Along with the common support condition

$$0 < \Pr(W = 1|X) < 1, \quad 13.$$

Equation 12 allows identification of treatment effect parameters. Equation 12 implies

$$E[Y_1 - Y_0|X] = E[Y|X, W = 1] - E[Y|X, W = 0].$$

Then, the common support condition in Equation 13 implies

$$\tau_{ATE} = E[E[Y|X, W = 1] - E[Y|X, W = 0]] \quad 14.$$

and

$$\tau_{ATE} = E[E[Y|X, W = 1] - E[Y|X, W = 0]|W = 1]. \quad 15.$$

Other causal parameters of interest, like the quantile treatment effects in Equation 8, are identified by the combination of unconfoundedness and common support (Firpo 2007).

The common support condition can be directly assessed in the data, as it depends on the conditional distribution of X given W , which is identified by the joint distribution of (Y, W, X) . Unconfoundedness, however, is harder to assess, as it depends on potential outcomes that are not always observed. Pearl (2009) and others have investigated graphical causal structures that allow identification by conditioning on covariates. In particular, Pearl's backdoor criterion (Pearl 1993) provides sufficient conditions under which treatment effect parameters are identified via conditioning.⁶

In our college financial aid example, suppose that other causes of the award, aside from the college entry exam, do not have a direct effect on college grades (that is, all the effect of V on Y is through W). This is the setting in **Figure 3b**.⁷ Using Pearl's backdoor criterion, and provided that a common support condition holds, it can be shown that the causal structure in **Figure 3b**

⁴For the identification results stated below, it is enough to have unconfoundedness in terms of the marginal distributions of potential outcomes, that is, $Y_w \perp\!\!\!\perp W|X$ for $w = \{0, 1\}$.

⁵The unconfoundedness assumption in Equation 12 is often referred to as conditional independence, exogeneity, selection on observables, ignorability, or missing at random.

⁶Pearl (2009) and Morgan & Winship (2015) provide detailed introductions to identification in DAGs, and Richardson & Robins (2013) discuss the relationship between identification in DAGs and statements about conditional independence between the treatment and potential outcomes similar to that in Equation 12.

⁷Notice that the node representing V disappears from the DAG in **Figure 3b**. Because V causes only one variable in the DAG, it cannot create confounding. As a result, it can be safely excluded from the DAG.

implies that treatment effect parameters are identified by adjusting for X , as in Equations 14 and 15. The intuition for this result is rather immediate. Precollege ability, U , is a common cause of W and Y , and the only confounder in this DAG. However, conditioning on X blocks the path from U to X . Once we condition on X , the variable U is no longer a confounder, as it is not a cause of W . In contrast, if other causes of the award, aside from the college entry exam, have a direct effect on college grades, as in **Figure 3a**, then Pearl's backdoor criterion does not imply Equations 14 or 15. Additional confounding, created by V , is not controlled for by conditioning on X alone.

4.2. Regression Adjustments

There exists a wide array of methods for estimation and inference under unconfoundedness. However, it is important to notice that, in the absence of additional assumptions, least squares regression coefficients do not identify ATE or ATET. Consider a simple setting where X takes on a finite number of values $x_1, \dots, x_m$. Under Equations 12 and 13, ATE and ATET are given by

$$\tau_{\text{ATE}} = \sum_{k=1}^m \left(E[Y|X = x_k, W = 1] - E[Y|X = x_k, W = 0] \right) \Pr(X = x_k)$$

and

$$\tau_{\text{ATET}} = \sum_{k=1}^m \left(E[Y|X = x_k, W = 1] - E[Y|X = x_k, W = 0] \right) \Pr(X = x_k | W = 1),$$

respectively. These identification results form the basis for the subclassification estimators of ATE and ATET in the work of Cochran (1968) and Rubin (1977). One could, however, attempt to estimate ATE or ATET using least squares. In particular, one could consider estimating τ_{OLS} by least squares in the regression equation

$$Y_i = \tau_{\text{OLS}} W_i + \sum_{k=1}^m \beta_{\text{OLS},k} D_{ki} + \varepsilon_i,$$

under the usual restrictions on ε_i , where D_{ki} is a binary variable that takes value one if $X_i = x_k$ and zero otherwise. It can be shown that, in the absence of additional restrictive assumptions, τ_{OLS} differs from τ_{ATE} and τ_{ATET} . More precisely, it can be shown that

$$\tau_{\text{OLS}} = \sum_{k=1}^m (E[Y|X = x_k, W = 1] - E[Y|X = x_k, W = 0]) w_k$$

where

$$w_k = \frac{\text{var}(W|X = x_k) \Pr(X = x_k)}{\sum_{r=1}^m \text{var}(W|X = x_r) \Pr(X = x_r)}.$$

That is, τ_{OLS} identifies a variance-weighted ATE (see, e.g., Angrist & Pischke 2008).⁸ Even in this simple setting, with a regression specification that is fully saturated in all values of X (including a dummy variable for each possible value), τ_{OLS} differs from τ_{ATE} and τ_{ATET} except for special cases (e.g., when $E[Y|X = x_k, W = 1] - E[Y|X = x_k, W = 0]$ is the same for all k).

⁸It can be shown that $\text{var}(W|X = x_r)$ is maximal when $\Pr(W = 1|X = x_r) = 1/2$ and is decreasing as this probability moves toward zero or one. That is, τ_{OLS} weights up groups with $X = x_k$, where the size of the treated and untreated populations are roughly equal, and weights down groups with large imbalances in the sizes of these two groups.

4.3. Matching Estimators

Partly because of the failure of linear regression to estimate conventional treatment effect parameters like ATE and ATET, researchers have resorted to flexible and nonparametric methods like matching.

4.3.1. Matching on covariates. Matching estimators of ATE and ATET can be constructed in the following manner. First, for each sample unit i , select a unit $j(i)$ in the opposite treatment group with similar covariate values. That is, select $j(i)$ such that $W_{j(i)} = 1 - W_i$ and $X_{j(i)} \simeq X_i$. Then, a one-to-one nearest-neighbor matching estimator of ATET can be obtained as the average of the differences in outcomes between the treated units and their matches,

$$\hat{\tau}_{\text{ATET}} = \frac{1}{n_1} \sum_{i=1}^n W_i (Y_i - Y_{j(i)}). \quad 16.$$

An estimator of ATE can be obtained in a similar manner, but using the matches for both treated and nontreated:

$$\hat{\tau}_{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n W_i (Y_i - Y_{j(i)}) - \frac{1}{n} \sum_{i=1}^n (1 - W_i) (Y_i - Y_{j(i)}). \quad 17.$$

Nearest-neighbor matching comes in many varieties. It can be done with replacement (that is, potentially using each unit in the control group as a match more than once) or without, with only one or several matches per unit (in which case, the outcomes of the matches for each unit are usually averaged before inserting them into Equations 16 and 17), and there exist several potential choices for the distance that measures the discrepancies in the values of the covariates between units [like the normalized Euclidean distance and the Mahalanobis distance (see Abadie & Imbens 2011, Imbens 2004)].

4.3.2. Matching on the propensity score. Matching on the propensity score is another common variety of matching method. Rosenbaum & Rubin (1983) define the propensity score as the conditional probability of receiving the treatment given covariate values

$$p(X) = \Pr(W = 1|X).$$

Rosenbaum & Rubin (1983) prove that, if the conditions in Equations 12 and 13 hold, then the same conditions hold after replacing X with the propensity score $p(X)$. An implication of this result is that, if controlling for X makes the treatment unconfounded, then controlling for $p(X)$ makes the treatment unconfounded, as well. **Figure 4** provides intuition for this result. The entire effect of X on W is mediated by the propensity score. Therefore, after conditioning on $p(X)$, the covariate X is no longer a confounder. In other words, conditioning on the propensity score blocks

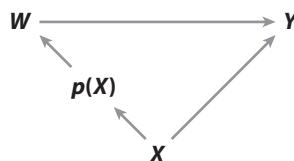


Figure 4

Identification by conditioning on the propensity score, $p(X)$. The graph shows the result of a college entrance exam, X ; other student characteristics affecting college admission, V ; receipt of a scholarship award, W ; and college grades, Y .

the backdoor path between W and Y . Therefore, any remaining statistical dependence between W and Y is reflective of the causal effect of W on Y . The propensity score result of Rosenbaum & Rubin (1983) motivates matching estimators that match on the propensity score, $p(X)$, instead of matching on the entire vector of covariates, X . Matching on the propensity score reduces the dimensionality of the matching variable, avoiding biases generated by curse of dimensionality issues (see, e.g., Abadie & Imbens 2006). In most practical settings, however, the propensity score is unknown. Nonparametric estimation of the propensity score brings back dimensionality issues. In some settings, estimation of the propensity score can be guided by institutional knowledge about the process that produces treatment assignment (elicited from experts in the subject matter). In empirical practice, the propensity score is typically estimated using parametric methods like probit or logit.

Abadie & Imbens (2006, 2011, 2012) derive large sample results for estimators that match directly on the covariates, X . Abadie & Imbens (2016) provide large sample results for the case where matching is done on an estimated propensity score (for reviews and further references on propensity score matching estimators, see Imbens & Rubin 2015, Rosenbaum 2010).

4.4. Inverse Probability Weighting

Inverse probability weighting (IPW) methods (see Hirano et al. 2003, Robins et al. 1994) are also based on the propensity score and provide an alternative to matching estimators. This approach proceeds by first obtaining estimates of propensity score values, $\hat{p}(X_i)$, and then using those estimates to weight outcome values. For example, an IPW estimator of ATE is

$$\hat{\tau}_{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n \frac{W_i Y_i}{\hat{p}(X_i)} - \frac{1}{n} \sum_{i=1}^n \frac{(1 - W_i) Y_i}{1 - \hat{p}(X_i)}.$$

This estimator uses $1/\hat{p}(X_i)$ to weigh observations in the treatment group and $1/(1 - \hat{p}(X_i))$ to weigh observations in the untreated group. Intuitively, observations with large $\hat{p}(X_i)$ are overrepresented in the treatment group and thus weighted down when treated. The same observations are weighted up when untreated. The opposite applies to observations with small $\hat{p}(X_i)$. This estimator can be modified so that the weights in each treatment arm sum to one, which produces improvements in finite sample performance (see, e.g., Busso et al. 2014).

4.5. Imputation and Projection Methods

Imputation and projection methods (e.g., Cattaneo & Farrell 2011, Heckman et al. 1998, Imbens et al. 2006, Little & Rubin 2002) provide an alternative class of estimators that rely on the assumptions in Equations 12 and 13. Regression imputation estimators are based on preliminary estimates of the outcome process, that is, the conditional distribution of Y given (X, W) . Let $\hat{\mu}_1(x)$ and $\hat{\mu}_0(x)$ be parametric or nonparametric estimators of $E[Y|X = x, W = 1]$ and $E[Y|X = x, W = 0]$, respectively. Then, an empirical analog of Equation 14 provides a projection and imputation estimator of ATE:

$$\hat{\tau}_{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n (\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)).$$

Similarly, a projection and imputation estimator of ATET is given by the empirical analog of Equation 15.

4.6. Hybrid Methods

Several hybrid methods combine matching and propensity score weighting with projection and imputation techniques. Among them are the doubly robust estimators (or, more generally, locally robust estimators) of Van der Laan & Robins (2003), Bang & Robins (2005), Cattaneo (2010), Farrell (2015), Chernozhukov et al. (2016), and Słoczyński & Wooldridge (2017). In the case of ATE estimation, for example, they can take the form

$$\begin{aligned}\hat{\tau}_{\text{ATE}} = & \frac{1}{n} \sum_{i=1}^n \left(\frac{W_i (Y_i - \hat{\mu}_1(X_i))}{\hat{p}(X_i)} + \hat{\mu}_1(X_i) \right) \\ & - \frac{1}{n} \sum_{i=1}^n \left(\frac{(1 - W_i) (Y_i - \hat{\mu}_0(X_i))}{1 - \hat{p}(X_i)} + \hat{\mu}_0(X_i) \right).\end{aligned}$$

This estimator is doubly robust in the sense that consistent estimation of $p(\cdot)$ or consistent estimation of $\mu_1(\cdot)$ and $\mu_0(\cdot)$ is required for the validity of the estimator. Part of the appeal of doubly and locally robust methods comes from the efficiency properties of the estimators. However, relative to other methods discussed in this section, doubly robust methods require estimating not only $p(\cdot)$ or $\mu_1(\cdot)$ and $\mu_0(\cdot)$, but all of these functions. As a result, relative to other methods, doubly robust estimators involve additional tuning parameters and implementation choices. Closely related to doubly and locally robust methods are the bias-corrected matching estimators of Abadie & Imbens (2011). These estimators include a regression-based bias correction designed to purge matching estimators from the bias that arises from imperfect matches in the covariates,

$$\begin{aligned}\hat{\tau}_{\text{ATE}} = & \frac{1}{n} \sum_{i=1}^n W_i \left((Y_i - Y_{j(i)}) - (\hat{\mu}_0(X_i) - \hat{\mu}_0(X_{j(i)})) \right) \\ & - \frac{1}{n} \sum_{i=1}^n (1 - W_i) \left((Y_i - Y_{j(i)}) - (\hat{\mu}_1(X_i) - \hat{\mu}_1(X_{j(i)})) \right).\end{aligned}$$

4.7. Comparisons of Estimators

Busso et al. (2014) study the finite sample performance of matching and IPW estimators, as well as some of their variants. Kang & Schafer (2007) and the accompanying discussions and rejoinder provide finite sample evidence on the performance of double robust estimators and other related methods.⁹

As these and other studies demonstrate, the relative performance of estimators based on conditioning on observables depends on the features of the data generating process. In finite samples, IPW estimators become unstable when the propensity score approaches zero or one, while regression imputation methods may suffer from extrapolation biases. Estimators that match directly on covariates do not require specification choices but may incorporate nontrivial biases if the quality of the matches is poor. Hybrid methods, such as bias-corrected matching estimators or doubly robust estimators, include safeguards against bias caused by imperfect matching and misspecification but impose additional specification choices that may affect the resulting estimate.

Apart from purely statistical properties, estimators based on conditioning on covariates differ in the way they are related to research transparency. In particular, during the design phase of a study (i.e., sample, variable, and model selection), matching and IPW methods employ only

⁹Dehejia & Wahba (1999), Smith & Todd (2005), and others evaluate the performance of some of the estimators in this section against an experimental benchmark.

information about treatment, W , and covariates, X . That is, the matches and the specification for the propensity score can be constructed without knowledge or use of the outcome variable. This feature of matching and IPW estimators provides a potential safeguard against specification searches and p -hacking, a point forcefully made by Rubin (2007).

4.8. Recent Developments and Additional Topics

All of the estimation and inference methods discussed above take the covariates, X , as known and small relative to the sample size, although, in practice, researchers often use high-dimensional models with the aim of using a setting where the conditional independence assumption is deemed appropriate. In most applications, X includes both raw preintervention covariates and transformations thereof, such as dummy variable expansions of categorical variables, power or other series expansions of continuous variables, higher-order interactions, or other technical regressors generated from the original available variables. The natural tension between standard theory (where the dimension of X is taken to be small) and practice (where the dimension of X tends to be large) has led to a new literature that aims to develop estimation and inference methods in program evaluation that account and allow for high-dimensional X .

Belloni et al. (2014) and Farrell (2015) develop new program evaluation methods employing machinery from the high-dimensional literature in statistics. This methodology, which allows for very large X (much larger than the sample size), proceeds in two steps. First, a parsimonious model is selected from the large set of preintervention covariates employing model selection via least absolute shrinkage and selection operator (LASSO). Then, treatment effect estimators are constructed using only the small (usually much smaller than the sample size) set of selected covariates. More general methods and a review of this literature are given by Belloni et al. (2017) and Athey et al. (2016), among others. Program evaluation methods proposed in this literature employ modern model selection techniques, which require a careful analysis and interpretation but often ultimately produce classical distributional approximations (in the sense that these asymptotic approximations do not change relative to results in low-dimensional settings).

Cattaneo et al. (2017a; 2018c,d) develop program evaluation methods where the distributional approximations do change relative to low-dimensional results due to the inclusion of many covariates in the estimation. These results can be understood as giving new distributional approximations that are robust to (i.e., also valid in) cases where either the researcher cannot select out covariates (e.g., when many multiway fixed effects are needed) or many covariates remain included in the model even after model selection. These high-dimensional methods not only are valid when many covariates are included, but also continue to be valid in cases where only a few covariates are used, thereby offering demonstrable improvements for estimation and inference in program evaluation. Related kernel-based methods are developed by Cattaneo & Jansson (2018).

All of the ideas and methods discussed above are mostly concerned with average treatment effects in the context of binary treatments. Many of these results have been extended to multivalued treatments (Cattaneo 2010, Farrell 2015, Hirano & Imbens 2004, Imai & van Dyk 2004, Imbens 2000, Lechner 2001, Yang et al. 2016), to quantiles and related treatment effects (Cattaneo 2010, Firpo 2007), and to the analysis and interpretation of counterfactual distributional treatment effects (Chernozhukov et al. 2013, DiNardo et al. 1996, Firpo & Pinto 2016). We do not discuss this work due to space limitations.

Space restrictions also prevent us from discussing in detail other strands of the literature on estimation of treatment effects based on conditioning on covariates. In particular, in the biostatistics literature, structural nested models, marginal structural models, and optimal treatment regimes estimators are employed to estimate treatment effects in contexts with time-varying treatments and

confounders (see, e.g., Lok et al. 2004, Murphy 2003, Robins 2000). Also, a large literature on sensitivity to unobserved confounders analyzes the impact of departures from the unconfoundedness assumption in Equation 12 (see, e.g., Altonji et al. 2005, Imbens 2003, Rosenbaum 2002).

5. DIFFERENCE IN DIFFERENCES AND SYNTHETIC CONTROLS

5.1. Difference in Differences

In the previous section, we consider settings where treatment assignment is confounded but where there exists a set of observed covariates, X , such that treatment assignment becomes unconfounded after conditioning on X . In many applied settings, however, researchers confront the problem of the possible existence of unobserved confounders. In these settings, difference-in-differences models aim to attain identification by restricting the way in which unobserved confounders affect the outcome of interest over time. Consider the following panel data regression model,

$$Y_{it} = W_{it}\tau_{it} + \mu_i + \delta_t + \varepsilon_{it},$$

where only Y_{it} and W_{it} are observed. We regard the mapping represented in this equation as structural (or causal). That is, the equation describes potential outcomes, which are now indexed by time period, t ,

$$\begin{aligned} Y_{0it} &= \mu_i + \delta_t + \varepsilon_{it}, \\ Y_{1it} &= \tau_{it} + \mu_i + \delta_t + \varepsilon_{it}. \end{aligned} \tag{18}$$

Then, we have $\tau_{it} = Y_{1it} - Y_{0it}$. To simplify the exposition, and because it is a common setting in empirical practice, we assume that there are only two time periods. Period $t = 0$ is the pretreatment period, before the treatment is available, so $W_{i0} = 0$ for all i . Period $t = 1$ is the posttreatment period, when a fraction of the population units are exposed to the treatment. δ_t is a time effect, common across units. We treat μ_i as a time invariant confounder, so μ_i and W_{i1} are not independent. In contrast, we assume that ε_{it} are causes of the outcome that are unrelated to selection for treatment, so $E[\varepsilon_{it}|W_{it}] = E[\varepsilon_{it}]$. This condition can be weakened to $E[\Delta\varepsilon_{i1}|W_{i1}] = E[\Delta\varepsilon_{i1}]$, where Δ is the first difference operator, so $\Delta\varepsilon_{i1} = \varepsilon_{i1} - \varepsilon_{i0}$. This type of structure is the same as the widespread linear fixed effect panel data model (see, e.g., Arellano 2003). Then, we have

$$\begin{aligned} E[Y_{i1}|W_{i1} = 1] &= E[\tau_{i1}|W_{i1} = 1] + E[\mu_i|W_{i1} = 1] + \delta_1 + E[\varepsilon_{i1}] \\ E[Y_{i0}|W_{i1} = 1] &= E[\mu_i|W_{i1} = 1] + \delta_0 + E[\varepsilon_{i0}] \\ E[Y_{i1}|W_{i1} = 0] &= E[\mu_i|W_{i1} = 0] + \delta_1 + E[\varepsilon_{i1}] \\ E[Y_{i0}|W_{i1} = 0] &= E[\mu_i|W_{i1} = 0] + \delta_0 + E[\varepsilon_{i0}]. \end{aligned}$$

In this model, the effect of the unobserved confounders on the average of the outcome variable is additive and does not change in time. It follows that

$$\begin{aligned} \tau_{\text{ATET}} &= E[\tau_{i1}|W_{i1} = 1] \\ &= \left[E[Y_{i1}|W_{i1} = 1] - E[Y_{i1}|W_{i1} = 0] \right] - \left[E[Y_{i0}|W_{i1} = 1] - E[Y_{i0}|W_{i1} = 0] \right] \\ &= \left[E[\Delta Y_{i1}|W_{i1} = 1] - E[\Delta Y_{i1}|W_{i1} = 0] \right]. \end{aligned} \tag{19}$$

The reason for the name difference in differences is apparent from Equation 19. Notice that τ_{ATET} is defined here for the posttreatment period, $t = 1$. Intuitively, identification in the difference-in-differences model comes from a common trends assumption implied by Equation 18. It can be

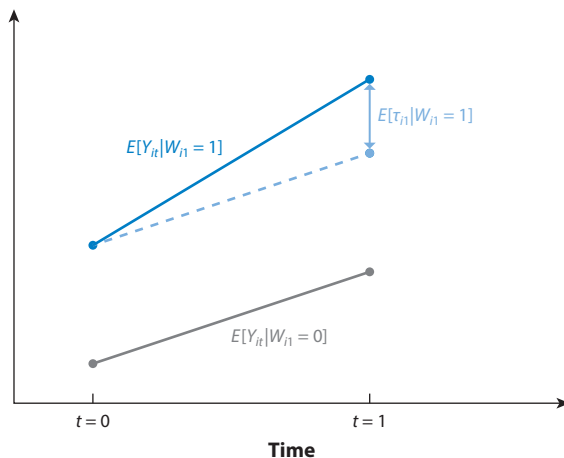


Figure 5

Identification in a difference-in-differences model. The dashed line represents the outcome that the treated units would have experienced in the absence of the treatment.

easily seen that Equation 18, along with $E[\Delta \varepsilon_{it}|W_{it}] = E[\Delta \varepsilon_{it}]$, implies

$$E[\Delta Y_{0i1}|W_{it} = 1] = E[\Delta Y_{0i1}|W_{it} = 0]. \quad 20.$$

That is, in the absence of the treatment, the average outcome for the treated and the average outcome for the nontreated would have experienced the same variation over time.

Figure 5 illustrates how identification works in a difference-in-differences model. In this setting, a set of untreated units identifies the average change that the outcome for the treated units would have experienced in the absence of the treatment. The set of untreated units selected to reproduce the counterfactual trajectory of the outcome for the treated is often called the control group in this literature (borrowing from the literature on randomized controlled trials).

Notice also that the common trend assumption in Equation 20 is not invariant to nonlinear transformations of the dependent variable. For example, if Equation 20 holds when the outcome is a wage rate measured in levels, then the same equation will not hold in general for wages measured in logs. In other words, identification in a difference-in-differences model is not invariant to nonlinear transformations in the dependent variable.

In a panel data regression (that is, in a setting with repeated pre- and posttreatment observations for the same units), the right-hand side of Equation 19 is equal to the regression coefficient on W_{i1} in a regression of ΔY_{i1} on W_{i1} and a constant. Consider, instead, a regression setting with pooled cross-sections for the outcome variable at $t = 0$ and $t = 1$ and information on W_{i1} for all of the units in each cross-section. In this setting, one cross-section contains information on (Y_{i0}, W_{i1}) and the other cross-section contains information on (Y_{i1}, W_{i1}) . Let T_i be equal to zero if unit i belongs to the pretreatment sample and equal to one if it belongs to the posttreatment sample. Then, the right-hand side of Equation 19 is equal to the coefficient on the interaction $W_{i1}T_i$ in a regression of the outcome on a constant, W_{i1} , T_i , and the interaction $W_{i1}T_i$ for a sample that pools the preintervention and postintervention cross-sections.

Several variations of the basic difference-in-differences model have been proposed in the literature. In particular, the model naturally extends to a fixed-effects regression in settings with more than two periods or models with unit-specific linear trends (see, e.g., Bertrand et al. 2004).

Logically, estimating models with many time periods or trends imposes greater demands on the data.

The common trends restriction in Equation 20 is clearly a strong assumption and one that should be assessed in empirical practice. The plausibility of this assumption can sometimes be evaluated using (a) multiple preintervention periods (like in Abadie & Dermisi 2008) or (b) population groups that are not at risk of being exposed to the treatment (like in Gruber 1994). In both cases, a test for common trends is based on the difference-in-differences estimate of the effect of a placebo intervention, that is, an intervention that did not happen. In the first case, the placebo estimate is computed using the preintervention data alone and evaluates the effect of a nonexistent intervention taking place before the actual intervention period. Rejection of a null effect of the placebo intervention provides direct evidence against common trends before the intervention. The second case is similar but uses an estimate obtained for a population group known not to be amenable to receiving the treatment. For example, Gruber (1994) uses difference in differences to evaluate the effect of the passage of mandatory maternity benefits in some US states on the wages of married women of childbearing age. In this context, Equation 20 means that, in the absence of the intervention, married women of childbearing age would have experienced the same increase in log wages in states that adopted mandated maternity benefits and states that did not. To evaluate the plausibility of this assumption, Gruber (1994) compares the changes in log wages in adopting and nonadopting states for single men and women over 40 years old.

The plausibility of Equation 20 may also be questioned if the treated and the control groups are different in the distribution of attributes that are known or suspected to affect the outcome trend. For example, if average earnings depend on age or on labor market experience in a nonlinear manner, then differences in the distributions of these two variables between treated and nontreated in the evaluation of an active labor market program pose a threat to the validity of the conventional difference-in-differences estimator. Abadie (2005b) proposes a generalization of the difference-in-differences model for the case when covariates explain the differences in the trends of the outcome variable between treated and nontreated. The resulting estimators adjust the distribution of the covariates between treated and nontreated using propensity score weighting.

Finally, Athey & Imbens (2006) provide a generalization of the difference-in-differences model to the case when Y_{0it} is nonlinear in the unobserved confounder, μ_i . Identification in this model comes from strict monotonicity of Y_{0it} with respect to μ_i and from the assumption that the distribution of μ_i is time invariant for the treated and the nontreated (although it might differ between the two groups). One of the advantages of their approach is that it provides an identification result that is robust to monotonic transformations of the outcome variable [e.g., levels, Y_{it} , versus logs, $\log(Y_{it})$].

5.2. Synthetic Controls

Difference-in-differences estimators are often used to evaluate the effects of events or interventions that affect entire aggregate units, such as states, school districts, or countries (see, e.g., Card 1990, Card & Krueger 1994). For example, Card (1990) estimates the effect of the Mariel Boatlift, a large and sudden influx of immigrants to Miami, on the labor market outcomes of native workers in Miami. In a difference-in-differences design, Card (1990) uses a combination of four other cities in the south of the United States (Atlanta, Los Angeles, Houston, and Tampa-St. Petersburg) to approximate the change in native unemployment rates that would have been observed in Miami in the absence of the Mariel Boatlift. Studies of this type are sometimes referred to as comparative case studies because a comparison group is selected to evaluate an instance or case of an event or policy of interest. In the work of Card (1990), the case of interest is the arrival of Cuban immigrants to

Miami during the Mariel Boatlift, and the comparison is provided by the combination of Atlanta, Los Angeles, Houston, and Tampa-St. Petersburg.

The synthetic control estimator of Abadie & Gardeazabal (2003) and Abadie et al. (2010, 2015) provides a data-driven procedure to create a comparison unit in comparative case studies. A synthetic control is a weighted average of untreated units chosen to reproduce characteristics of the treated unit before the intervention. The idea behind synthetic controls is that a combination of untreated units, as in the work of Card (1990), may provide a better comparison to the treatment unit than any untreated unit alone.

Synthetic controls are constructed as follows. For notational simplicity, we assume that there is only one treated unit. This is without loss of generality because, when there is more than one treated unit, the procedure described below can be applied for each treated unit separately. Suppose that we observe $J + 1$ units in periods $1, 2, \dots, T$. Unit 1 is exposed to the intervention of interest during periods $T_0 + 1, \dots, T$. The remaining J units are an untreated reservoir of potential controls (a donor pool). Let $w = (w_2, \dots, w_{J+1})'$ be a collection of weights, with $w_j \geq 0$ for $j = 2, \dots, J + 1$ and $w_2 + \dots + w_{J+1} = 1$. Each value of w represents a potential synthetic control. Let X_1 be a $(k \times 1)$ vector of preintervention characteristics for the treated unit. Similarly, let X_0 be a $(k \times J)$ matrix which contains the same variables for the unaffected units. The vector $w^* = (w_2^*, \dots, w_{J+1}^*)'$ is chosen to minimize $\|X_1 - X_0 w\|$, subject to the weight constraints. The synthetic control estimator of the effect of the treatment for the treated unit in a postintervention period t ($t \geq T_0$) is

$$\hat{\alpha}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt}.$$

A weighted Euclidean norm is commonly employed to measure the discrepancy between the characteristics of the treated unit and the characteristics of the synthetic control

$$\|X_1 - X_0 w\| = \sqrt{(X_1 - X_0 w)' V (X_1 - X_0 w)},$$

where V is a diagonal matrix with nonnegative elements in the main diagonal that control the relative importance of obtaining a good match between each value in X_1 and the corresponding value in $X_0 w^*$. Abadie & Gardeazabal (2003) and Abadie et al. (2010, 2015) propose data-driven selectors of V . Abadie et al. (2010) propose an inferential method for synthetic controls based on Fisher's randomization inference.

It can be shown that, in general (that is, ruling out certain degenerate cases), if X_1 does not belong to the convex hull of the columns of X_0 , then w^* is unique and sparse. Sparsity means that w^* has only a few nonzero elements. However, in some applications, especially in applications with a large number of treated units, there may be synthetic controls that are not unique or sparse. Abadie & L'Hour (2017) propose a variant of the synthetic control estimator that addresses this issue. Their estimator minimizes

$$\|X_1 - X_0 w\|^2 + \lambda \sum_{j=2}^{J+1} w_j \|X_j - X_1\|^2,$$

with $\lambda > 0$, subject to the weight constraints. This estimator includes a penalization for pairwise matching discrepancies between the treated unit and each of the units that contribute to the synthetic control. Abadie & L'Hour (2017) show that, aside from special cases, if $\lambda > 0$, then the synthetic control estimator is unique and sparse. Hsiao et al. (2012) and Doudchenko & Imbens (2016) propose variants of the synthetic control framework that extrapolate outside of the support of the columns of X_0 . Recent strands of the literature extend synthetic control estimators using

matrix completion techniques (see Amjad et al. 2017, Athey et al. 2017) and propose sampling-based inferential methods (Hahn & Shi 2017).

6. INSTRUMENTAL VARIABLES

6.1. The Framework

Instrumental variables (IV) methods are widely used in program evaluation. They are easiest to motivate in the context of a randomized experiment with imperfect compliance, that is, a setting where experimental units may fail to comply with the randomized assignment of the treatment. Other settings where IV methods are used include natural experiments in observation studies; measurement error; nonlinear models with endogeneity; and dynamic, panel data, or time series contexts (for an overview and further references, including material on weak and many instruments, see, e.g., Imbens 2014, Wooldridge 2010).

Imperfect compliance with a randomized assignment is commonplace in economics and other disciplines where assignment affects human subjects, so compliance may be difficult to enforce or not desirable (for example, in active labor market programs for the unemployed, where the employment status of the individuals participating in the experiment may change between randomization and treatment). In settings where the experimenter is the only possible provider of the treatment, randomized assignment can be enforced in the control group. Such a setting is often referred to as one-sided noncompliance.

In randomized experiments with noncompliance, randomization of treatment intake fails because of postrandomization selection into the treatment. Assignment itself remains randomized, and the average effect of assignment on an observed outcome is identified. Assignment effects are often referred to as intention-to-treat effects. In some settings, the most relevant policy decision is whether or not to provide individuals with access to treatment, so intention-to-treat effects have direct policy relevance. In other instances, researchers and policy makers are interested in the effects of treatment intake, rather than on the effects of eligibility for treatment. In such cases, IV methods can be used to identify certain treatment effects parameters.

IV methods rely on the availability of instruments, that is, variables that affect treatment intake but do not directly affect potential outcomes. In the context of randomized experiments with imperfect compliance, instruments are usually provided by randomized assignment to treatment. We use Z to represent randomized assignment, so $Z = 1$ indicates assignment to the treatment group, and $Z = 0$ indicates assignment to the control group. As in previous sections, actual treatment intake is represented by the binary variable, W .

Figure 6 offers a DAG representation of the basic IV setting. The treatment, W , is affected directly by an unobservable confounder, U , which also directly affects the outcome variable, Y . Because U is unobserved, conditioning on U is unfeasible. However, in this case, there is an instrument, Z , that affects treatment, W , and only affects the outcome, Y , through W .

The key idea underlying IV methods is that the exogenous variation in the instrument, Z , which induces changes in the treatment variable, W , can be used to identify certain parameters of



Figure 6

Instrumental variable in a directed acyclic graph. The graph shows the treatment, W ; an unobservable confounder, U ; the outcome variable, Y ; and an instrument, Z , affecting the treatment.

interest in program evaluation and causal inference. Variation in Z is exogenous (or unconfounded) because Z affects Y only through W and because there are no common causes of Z and Y . Chalakh & White (2011) provide a detailed account of identification procedures via IV.

6.2. Local Average Treatment Effects

Employing a potential outcomes framework, Imbens & Angrist (1994) and Angrist et al. (1996) provide a formal setting to study identification and estimation of treatment effects under imperfect compliance. The observed treatment status is

$$W = ZW_1 + (1 - Z)W_0,$$

where W_1 and W_0 denote the potential treatment status under treatment and control assignment, respectively. Perfect compliance corresponds to $\Pr(W_1 = 1) = \Pr(W_0 = 0) = 1$, so $\Pr(W = Z) = 1$. One-sided noncompliance corresponds to the case when $\Pr(W_0 = 0) = 1$ but $\Pr(W_1 = 1) < 1$. In this case, units assigned to the control group never take the treatment, while units assigned to the treatment group may fail to take the treatment.

More generally, in the absence of perfect or one-sided compliance, the population can be partitioned into four groups defined in terms of the values of W_1 and W_0 . Angrist et al. (1996) define these groups as compliers ($W_1 > W_0$ or $W_0 = 0$ and $W_1 = 1$), always-takers ($W_1 = W_0 = 1$), never-takers ($W_1 = W_0 = 0$), and defiers [$W_1 < W_0$ ($W_0 = 1$ and $W_1 = 0$)]. Experimental units that comply with treatment assignment in both treatment arms are called compliers. Always-takers and never-takers are not affected by assignment. Defiers are affected by assignment in the opposite direction as expected: They take the treatment when they are assigned to the control group but refuse to take it when they are assigned to the treatment group.

We assume that treatment assignment is not trivial (in the sense that assignment is not always to the treatment group or always to the control group). In addition, we assume that the assignment has an effect on treatment intake:

$$0 < \Pr(Z = 1) < 1 \quad \text{and} \quad \Pr(W_1 = 1) \neq \Pr(W_0 = 1). \quad 21.$$

A key assumption in the IV setting is that potential outcomes, Y_1 and Y_0 , do not depend on the value of the instrument. This is called the exclusion restriction. In **Figure 6**, the exclusion restriction is manifested in that the effect of Z on Y is completely mediated by W . The combination of the exclusion restriction with the assumption that the instrument is randomized (or as good as randomized) implies

$$(Y_1, Y_0, W_1, W_0) \perp\!\!\!\perp Z. \quad 22.$$

In the language of IV estimation of linear regression models with constant coefficients, Equations 21 and 22 amount to the assumptions of instrument relevance and instrument exogeneity, respectively (see, e.g., Stock & Watson 2003).

Consider the IV parameter

$$\tau_{IV} = \frac{\text{cov}(Y, Z)}{\text{cov}(D, Z)} = \frac{E[Y|Z = 1] - E[Y|Z = 0]}{E[W|Z = 1] - E[W|Z = 0]}. \quad 23.$$

Under the assumptions in Equations 21 and 22, it can be shown that

$$\tau_{IV} = \frac{E[(Y_1 - Y_0)(W_1 - W_0)]}{E[W_1 - W_0]}. \quad 24.$$

It follows that, if the treatment does not vary across units, that is, if $Y_1 - Y_0$ is constant, τ_{IV} is equal to the constant treatment effect.¹⁰ The sample analog of τ_{IV} ,

$$\hat{\tau}_{IV} = \frac{\left(\sum_{i=1}^n Z_i Y_i / \sum_{i=1}^n Z_i \right) - \left[\sum_{i=1}^n (1 - Z_i) Y_i / \sum_{i=1}^n (1 - Z_i) \right]}{\left(\sum_{i=1}^n Z_i W_i / \sum_{i=1}^n Z_i \right) - \left[\sum_{i=1}^n (1 - Z_i) W_i / \sum_{i=1}^n (1 - Z_i) \right]},$$

provides an estimator of the constant treatment effect. This is often called the Wald estimator.

In the presence of treatment effect heterogeneity, however, it can be easily shown that τ_{IV} may fail to identify an average treatment effect for a well-defined population. In particular, it is easy to construct examples where $Y_1 - Y_0 > 0$ with probability one but $\tau_{IV} = 0$. The reason is that positive values of $(Y_1 - Y_0)(W_1 - W_0)$ given by positive values of $W_1 - W_0$ may cancel out with negative values in the expectation in the numerator of the expression for τ_{IV} in Equation 24.

Imbens & Angrist (1994) define the local average treatment effect (LATE) as

$$\tau_{LATE} = E[Y_1 - Y_0 | W_1 > W_0].$$

That is, in the terminology of Angrist et al. (1996), LATE is the average effect of the treatment for compliers, or the average effect of the treatment for those units that would always be in compliance with the assignment no matter whether they are assigned to the treatment or to the control group.

Suppose that the assumptions in Equations 21 and 22 hold. Suppose also that

$$\Pr(W_1 \geq W_0) = 1. \tag{25}$$

The assumption in Equation 25 is often referred to as monotonicity, and it rules out the existence of defiers. Then, we have

$$\tau_{IV} = \tau_{LATE}. \tag{26}$$

Monotonicity implies that $W_1 - W_0$ is binary, so the result in Equation 26 follows easily from Equation 24.

LATE represents average treatment effect for the units that change their treatment status according to their treatment assignment. Notice that, in the absence of additional restrictions, compliers are not individually identified because only W_1 or W_0 , but not both, are observed in practice. However, the result in Equation 26 implies that average treatment effects are identified for compliers.

A key identifying assumption for LATE is the monotonicity condition (Equation 25), which rules out defiers. Balke & Pearl (1997), Imbens & Rubin (1997), Heckman & Vytlačil (2005), and Kitagawa (2015) discuss testable implications of this assumption.

One-sided noncompliance, $\Pr(W_0 = 1) = 0$, is fairly common in settings where the experimenter is the only potential provider of the treatment. It has important practical implications. In particular, one-sided noncompliance implies monotonicity. It also implies that LATE is equal to ATET. That is, under one-sided noncompliance, we obtain

$$\tau_{LATE} = \tau_{ATET}.$$

More generally, however, LATE is not the same as ATE or ATET, and the interest in and interpretation of the LATE parameter has to be judged depending on the application at hand.

¹⁰More generally, if $Y_1 - Y_0$ is mean independent of $W_1 - W_0$, then τ_{IV} is equal to ATE.

Suppose, for example, that granting access to treatment represents a policy option of interest. Then, in a randomized experiment with imperfect compliance (and if monotonicity holds), the LATE parameter recovers the effect of the treatment for those units affected by being granted access to treatment, which is the policy decision under consideration (for further discussion, see, e.g., Imbens 2010).

Angrist et al. (2000) extend the methods described in this section to estimation in simultaneous equation models. This work connects with the marginal treatment effect (MTE) estimators discussed in Section 6.4. The local nature of LATE—that is, the fact that it applies only to compliers—raises the issue of under what conditions LATE estimates can be generalized to larger or different populations (for a discussion of extrapolation of LATE estimates and related problems, see Angrist & Fernandez-Val 2013).

6.3. General Identification and Estimation Results for Compliers (Kappa)

Abadie (2003) proposes a general class of estimators of LATE-type parameters (that is, parameters defined for the population of compliers). Like in the work of Imbens & Angrist (1994) and Angrist et al. (1996), Abadie (2003) adopts a context akin to a randomized experiment with imperfect compliance, where the instrument and treatment are binary, extending it to the case where conditioning on covariates may be needed for instrument validity. In particular, in the absence of unconditional randomization of the treatment, there may be common causes, X , that affect the instrument and the outcome. In this case, the instrument will not be valid unless validity is assessed after conditioning on X .

Consider versions of Equations 21, 22, and 25, where the same assumptions hold after conditioning on X . Define

$$\kappa_0 = (1 - W) \frac{(1 - Z) - \Pr(Z = 0|X)}{\Pr(Z = 0|X) \Pr(Z = 1|X)},$$

$$\kappa_1 = W \frac{Z - \Pr(Z = 1|X)}{\Pr(Z = 0|X) \Pr(Z = 1|X)},$$

$$\kappa = 1 - \frac{W(1 - Z)}{\Pr(Z = 0|X)} - \frac{(1 - W)Z}{\Pr(Z = 1|X)}.$$

Abadie (2003) shows that

$$E[b(Y_0, X)|W_1 > W_0] = \frac{1}{\Pr(W_1 > W_0)} E[\kappa_0 b(Y, X)], \quad 27.$$

$$E[b(Y_1, X)|W_1 > W_0] = \frac{1}{\Pr(W_1 > W_0)} E[\kappa_1 b(Y, X)], \quad 28.$$

and

$$E[g(Y, W, X)|W_1 > W_0] = \frac{1}{\Pr(W_1 > W_0)} E[\kappa g(Y, W, X)], \quad 29.$$

where b and g are arbitrary functions (provided that the expectations exist), and

$$\Pr(W_1 > W_0) = E[\kappa_0] = E[\kappa_1] = E[\kappa].$$

Equations 27–29 provide a general identification result for compliers. In particular, Equations 27–29 identify the distribution of attributes, X , for compliers, e.g.,

$$E[X|W_1 > W_0] = E[\kappa X]/E[\kappa].$$

This result allows a more detailed interpretation of the LATE parameter of Imbens & Angrist (1994) and Angrist et al. (1996). Although, as indicated above, compliers are not identified individually, the distribution of any observed characteristic can be described for the population of compliers. More generally, Equations 27–29 allow identification of regression models for compliers. Such models can be used to describe how the average effect of the treatment for compliers changes with X (for details, see Abadie 2003). Equations 27 and 28 can be applied to the identification of average complier responses to treatment or no treatment, as well as identification of the ATE for compliers. In particular, making $b(Y, X) = Y$, we obtain

$$E[Y_1 | W_1 > W_0] = \frac{E[\kappa_1 Y]}{E[\kappa_1]},$$

$$E[Y_0 | W_1 > W_0] = \frac{E[\kappa_0 Y]}{E[\kappa_0]}.$$

Therefore, we have

$$\begin{aligned}\tau_{\text{LATE}} &= \frac{E[\kappa_1 Y]}{E[\kappa_1]} - \frac{E[\kappa_0 Y]}{E[\kappa_0]} \\ &= \frac{E\left[Y \frac{Z - \Pr(Z = 1|X)}{\Pr(Z = 0|X) \Pr(Z = 1|X)}\right]}{E\left[W \frac{Z - \Pr(Z = 1|X)}{\Pr(Z = 0|X) \Pr(Z = 1|X)}\right]}.\end{aligned}$$

Kappa-based estimators are sample analogs of the identification results in this section. They require first-step estimation of $\Pr(Z = 1|X)$. For the same setting, Frölich (2007) derives an alternative estimator of τ_{LATE} based on the sample analog of the identification result

$$\tau_{\text{LATE}} = \frac{E[E[Y|X, Z = 1] - E[Y|X, Z = 0]]}{E[\Pr(W = 1|X, Z = 1) - \Pr(W = 1|X, Z = 0)]}.$$

An estimator of τ_{LATE} based on this result requires first-step estimation of $E[Y|X, Z = 1]$, $E[Y|X, Z = 0]$, $\Pr(W = 1|X, Z = 1)$, and $\Pr(W = 1|X, Z = 0)$.

Hong & Nekipelov (2010) derive efficient instruments for kappa-based estimators. Ogburn et al. (2015) propose doubly robust estimators of LATE-like parameters. Abadie (2002), Abadie et al. (2002), and Frölich & Melly (2013) propose estimators of distributional effects for compliers. Chernozhukov & Hansen (2005, 2013) propose a rank similarity condition under which they derive an estimator of quantile treatment effects for the entire population (rather than for compliers alone).

6.4. Marginal Treatment Effects

A closely related causal framework for understanding IV (and other program evaluation) settings was introduced by Björklund & Moffitt (1987); later developed and popularized by Heckman & Vytlacil (1999, 2005) and Heckman et al. (2006); and more recently studied by Kline & Walters (2016) and Brinch et al. (2017), among others. In this approach, the main parameters of interest are the MTEs and certain functionals thereof.

To describe the model and MTE estimand, we continue to employ potential outcomes but specialize the notation. In this setting, the instrument, Z , may be a vector of discrete or continuous random variables, and the observed data are assumed to be generated according to the following model. Let X be a vector of covariates. Conditional on $X = x$, we have

$$W = 1(p(Z) \geq V), \quad V|Z \sim \text{Uniform}[0, 1], \quad 30.$$

$$Y_1 = \mu_1(x) + U_1, \quad Y_0 = \mu_0(x) + U_0, \quad 31.$$

where $\mu_1()$ and $\mu_0()$ are unknown functions, $1(A)$ is the indicator function that takes value one if A occurs and value zero otherwise, and (U_1, U_0, V) have mean zero and are independent of Z (given $X = x$). The condition $V|Z \sim \text{Uniform}[0, 1]$ can be obtained as a normalization provided that V has an absolutely continuous distribution. Equation 30 implies $p(Z) = \Pr(W = 1|Z)$, the propensity score, which is, in this setting, a function of Z . For expositional simplicity, we assume that Z includes all elements of X .

Conditional on $X = x$, the MTE at level v is defined as

$$\tau_{\text{MTE}}(v|x) = E[Y_1 - Y_0|X = x, V = v].$$

The parameter $\tau_{\text{MTE}}(v|x)$ is understood as the treatment effect for an individual who is at the margin of being treated when $p(Z) = v$ and $X = x$. From a conceptual perspective, the MTE is useful because other treatment and policy effects can be recovered as weighted integrals of the MTE, where the weighting scheme depends on the parameter of interest (e.g., τ_{ATE} , τ_{ATET} , or τ_{LATE}). From a practical perspective, the MTE can be used to conduct certain counterfactual policy evaluations, sometimes going beyond what can be learned from the more standard treatment effect parameters in the IV literature, provided additional assumptions hold (for more details and discussion, see Heckman & Vytlačil 1999, 2005).

The key identifying assumptions for the MTE and related parameters are that, conditional on $X = x$, Z is nondegenerate and $(U_1, U_0, V) \perp\!\!\!\perp Z$, together with the common support condition $0 < p(Z) < 1$. Define the local IV (LIV) parameter as

$$\tau_{\text{LIV}}(p|x) = \frac{\partial}{\partial p} E[Y|X = x, p(Z) = p].$$

The parameter τ_{LIV} can be seen as a limiting version of LATE. For a continuous $p(Z)$, let

$$\Delta(p, b, x) = \frac{E[Y|X = x, p(Z) = p + b] - E[Y|X = x, p(Z) = p - b]}{E[W|X = x, p(Z) = p + b] - E[W|X = x, p(Z) = p - b]}, \quad 32.$$

where $E[W|X = x, p(Z) = p + b] - E[W|X = x, p(Z) = p - b] = 2b$. By the LATE result, it follows that

$$\Delta(p, b, x) = E[Y_1 - Y_0|X = x, p - b < V \leq p + b]. \quad 33.$$

Consider any x in the support of X and any limit point p of the support of $p(Z)|X = x$. Taking limits as $b \rightarrow 0$, on the right-hand sides of Equations 32 and 33, we have

$$\tau_{\text{MTE}}(p|x) = \tau_{\text{LIV}}(p|x).$$

As a result, the MTE is nonparametrically identified by the LIV parameter.

In practice, because of the difficulties associated with nonparametric estimations of functions, researchers often employ parametric forms $E[Y|X = x, p(Z) = p] = m(x, p, \theta)$ for the estimation of the MTE and functionals thereof. Then, the MTE is estimated as $\widehat{\tau}_{\text{MTE}}(p|x) = \partial m(x, p, \widehat{\theta})/\partial p$, where $\widehat{\theta}$ is an estimator of θ , and $p(Z)$ is estimated in a first step.

An alternative approach to LIV estimation of the MTE is given by a control function representation of the estimand (for a review of control function methods in econometrics, see Wooldridge 2015). This produces a two-step estimator of the MTE but with two regression functions, $E[Y|X = x, p(Z) = p, W = 1]$ and $E[Y|X = x, p(Z) = p, W = 0]$, estimated separately [both depending on the first-step estimate of $p(Z)$]. For recent examples and further

developments, including the study of cases with discrete instruments, the reader is referred to Kline & Walters (2016) and Brinch et al. (2017).

7. THE REGRESSION DISCONTINUITY DESIGN

The regression discontinuity (RD) design allows researchers to learn about causal effects in settings where the treatment is not explicitly randomized and unobservables cannot be ruled out. In its most basic form, this design is applicable when each unit receives a score, also called a running variable or index, and only units whose scores are above a known cutoff point are assigned to treatment status, while the rest are assigned to control status. Examples include antipoverty social programs assigned on a need basis after ranking units based on a poverty index and educational programs assigned on a performance basis after ranking units based on a test score (for early reviews, see Imbens & Lemieux 2008, Lee & Lemieux 2010; for an edited volume with a recent overview, see Cattaneo & Escanciano 2017; for a practical introduction, see Cattaneo et al. 2018a,b).

The RD treatment assignment mechanism is $W = 1(X \geq c)$, where X denotes the score, and c denotes the cutoff point. The key idea underlying any RD design is that units near the cutoff are comparable. At the core of the design is the assumption that units are not able to precisely manipulate their score to systematically place themselves above or below the cutoff. The absence of such endogenous sorting into treatment and control status is a crucial identifying assumption in the RD framework.

There is a wide variety of RD designs and closely related frameworks. Sharp RD designs are settings where treatment assignment, as determined by the value of the running variable relative to the cutoff, and actual treatment status coincide. This is akin to perfect compliance in randomized experiments. Fuzzy RD designs are settings where compliance with treatment assignment, as determined by the value of the running variable, is imperfect. This is analogous to the IV setting but for units with values of the running variable close to the cutoff. Kink RD designs are sharp or fuzzy settings where the object of interest is the derivative, rather than the level of the average outcome near or at the cutoff (see Card et al. 2015). Multicutoff RD designs are settings where the treatment assignment rule depends on more than one cutoff point (see Cattaneo et al. 2016). Multiscore and geographic RD designs are settings where the treatment assignment rule depends on more than one score variable (see Keele & Titiunik 2015, Papay et al. 2011).

In the canonical sharp RD design with a continuously distributed score, the most common parameter of interest is

$$\tau_{\text{SRD}} = E[Y_1 - Y_0 | X = c] = \lim_{x \downarrow c} E[Y | X = x] - \lim_{x \uparrow c} E[Y | X = x], \quad 34.$$

which corresponds to the average causal treatment effect at the score level $X = c$. **Figure 7** provides a graphical representation of the RD design. If the score variable satisfies $X < c$ then units are assigned to the control group ($W = 0$) and $E[Y_0 | X = x]$ is observed, while if $X \geq c$ then units are assigned to the treatment group ($W = 1$) and $E[Y_1 | X = x]$ is observed. The parameter of interest is the jump at the point $X = c$ where treatment assignment changes discontinuously.

Underlying the second equality in Equation 34 is the key notion of comparability between units with very similar values of the score but on opposite sides of the cutoff. This idea dates back to the original landmark contribution of Thistlethwaite & Campbell (1960) but was formalized more recently from a continuity-at-the-cutoff perspective by Hahn et al. (2001). To be specific, under regularity conditions, if the regression functions $E[Y_1 | X = x]$ and $E[Y_0 | X = x]$ are continuous at $x = c$ then τ_{SRD} is nonparametrically identified. The continuity conditions imply that the average response of units just below the cutoff provides a close approximation to the average response that would have been observed for units just above the cutoff had they not been assigned to treatment.

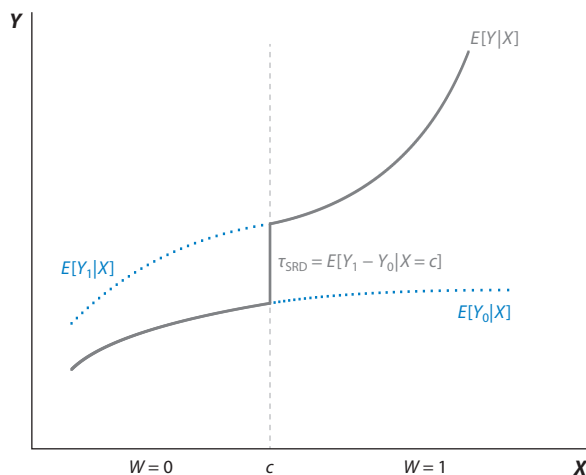


Figure 7

Regression discontinuity design. The outcome of interest is Y ; the score is X , and the cutoff point is c . Treatment assignment and status is $W = 1(X \geq c)$. The graph includes the estimable regression functions (solid lines) and their corresponding unobservable portions (dotted lines).

7.1. Falsification and Validation

An important feature of all RD designs is that their key identifying assumption can be supported by a variety of empirical tests, which enhances the plausibility of the design in applications. The most popular approaches for validating RD designs in practice include (a) testing for balance on preintervention covariates among units near the cutoff (Lee 2008), (b) testing for continuity of the score's density function near the cutoff (Cattaneo et al. 2017a, McCrary 2008), and (c) graphing an estimate of $E[Y|X = x]$ at the cutoff and away from the cutoff (Calonico et al. 2015). Other falsification and validation methods are also used in practice, although these are usually tailored to special cases (for a practical introduction to these methods, see Cattaneo et al. 2018a).

7.2. Estimation and Inference

The continuity assumptions on conditional expectations at the cutoff, which necessarily rely on the score variable being continuously distributed near the cutoff, imply that extrapolation is unavoidable in canonical RD designs. Observations near the cutoff are used to learn about the fundamentally unobserved features of the conditional expectations $E[Y_1|X = c]$ and $E[Y_0|X = c]$ at the cutoff. Local polynomial regression methods are the most common estimation approaches in this context, as they provide a good compromise between flexibility and practicality.

The core idea underlying local polynomial estimation and inference involves three simple steps: (a) localize the sample near the cutoff by discarding observations with scores far from the cutoff, (b) employ weighted least-squares to approximate the regression functions on either side of the cutoff separately, and (c) predict and extrapolate the value of the regression functions at the cutoff. More formally, assuming a random sample of size n , the standard local-linear RD estimator $\hat{\tau}_{\text{SRD}}$ is obtained by the local (to the cutoff c) weighted linear least squares regression

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\tau}_{\text{SRD}} \\ \hat{\beta} \end{bmatrix} = \arg \min_{\beta_0, \tau, \beta_1, \beta_2} \sum_{i=1}^n \left[Y_i - \beta_0 - W_i \tau - (X_i - c) \beta_1 - (X_i - c) W_i \beta_2 \right]^2 K \left(\frac{X_i - \bar{x}}{b} \right),$$

where $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2)'$, $K(\cdot)$ denotes a compact-supported (kernel) weighting function, and h denotes the bandwidth around the cutoff c . Thus, this is a local regression that employs only observations $X_i \in [c - h, c + h]$ with weights according to the kernel function; the two more common weighting schemes are equal weights (uniform kernel) and linear decreasing weights (triangular kernel). The coefficient $\hat{\beta}_0$ is an estimator of $E[Y_0|X = c]$ and can be used to assess the economic significance of the RD treatment effect estimate $\hat{\tau}_{\text{SRD}}$. The coefficients $\hat{\beta}$ capture the nonconstant, linear relationship between the outcome and score variables on either side of the cutoff.

Localization near the cutoff via the choice of bandwidth h is crucial and should be implemented in a systematic and objective way; ad hoc bandwidth selection methods can hamper empirical work using RD designs. Principled methods for bandwidth selection have been recently developed (for an overview, see Cattaneo & Vazquez-Bare 2016). Imbens & Kalyanaraman (2012) develop mean-squared-error (MSE) optimal bandwidth selection and RD point estimation, while Calonico et al. (2014) propose a robust bias-correction approach leading to valid inference methods based on the MSE-optimal bandwidth and RD point estimator. Using higher-order distributional approximations, Calonico et al. (2018a,b) show that robust bias-corrected inference is demonstrably better than other alternatives and optimal in some cases. They also develop coverage optimal bandwidth selection methods to improve RD inference in applications. Calonico et al. (2018c) study inclusion of preintervention covariates using local polynomial methods in RD designs. A practical introduction to these methods is given by Cattaneo et al. (2018a,b), who also provide additional references.

While the continuity-based approach to RD designs is the most popular in practice, researchers often explicitly or implicitly rely on a local randomization interpretation of RD designs when it comes to the heuristic conceptualization of RD empirical methods and results (see, e.g., Lee 2008, Lee & Lemieux 2010). The two methodological frameworks are, however, substantively distinct because they generally require different assumptions (Sekhon & Titiunik 2016, 2017). At an intuitive level, the local randomization RD approach states that, for units with scores sufficiently close to the cutoff, the assignment mechanism resembles that in a randomized controlled trial. However, because the RD treatment assignment rule $W = 1(X \geq c)$ implies a fundamental lack of common support among control and treatment units, the local randomization RD framework requires additional assumptions for identification, estimation, inference, and falsification.

Cattaneo et al. (2015) introduce a Fisherian model for inference in RD designs using a local randomization assumption. The key assumption in this model is the existence of a neighborhood around the cutoff where two conditions hold: (a) The treatment assignment mechanism is known (e.g., complete randomization), and (b) the nonrandom potential outcomes are a function of the score only through treatment status (i.e., an exclusion restriction). Under these assumptions, they develop finite-sample exact randomization inference methods for the sharp null hypothesis of no treatment effect, construct confidence intervals under a given treatment effect model, and discuss several falsification methods. They also propose data-driven methods for neighborhood selection, based on the local randomization assumptions.

Cattaneo et al. (2017c) extend the basic Fisherian RD randomization inference framework, allowing for parametric adjustments of potential outcomes as functions of the score. This approach resembles the continuity-based approach and basic parametric regression adjustment methods. Cattaneo et al. (2017c) also discuss other approaches for RD analysis based on the analysis of experiments under a local randomization assumption and compare these methods to the continuity-based approach with local polynomial fitting. Relevant to the comparison between RD approaches is that the parameter of interest differs depending on the approach. In the continuity-based approach, τ_{SRD} is the only nonparametrically identified estimand. In the local randomization

approach, other parameters are identifiable and may be of equal or greater interest. In particular, a parameter of interest in some settings is

$$\tau_{\text{LR}}(b) = E[Y_1 - Y_0 | c - b \leq X \leq c + b].$$

τ_{LR} is the average treatment effect for units with scores falling within the local randomization window $[c - b, c + b]$ for some choice of window length $2b > 0$.

7.3. Extensions, Modifications, and Recent Topics

Another situation in which the distinction between parameters and methodological approaches is important is when the score variable is discrete. In this case, τ_{SRD} is not identified nonparametrically, which renders the basic continuity-based approach for identification inapplicable. Local polynomial fits can still be used if the score variable takes on many discrete values because this estimation method takes the number of mass points as the effective sample size. The discrete nature of the score, however, changes the interpretation of the canonical RD parameter, τ_{SRD} . In this setting, estimation and inference employs a fixed region or bandwidth around the cutoff, so identification relies on a local parametric extrapolation (see Lee & Card 2008).

Alternatively, when the score variable X is discrete (and more so when it has only a few mass points), it may be more natural to consider an alternative RD parameter:

$$\tau_{\text{DS}} = E[Y_1 | X = x_+] - E[Y_0 | X = x_-],$$

where x_+ and x_- denote the closest mass points above and below the cutoff c respectively (with either $x_+ = c$ or $x_- = c$). Under a local randomization assumption, τ_{DS} is nonparametrically identified, and estimation and inference can be conducted using any of the methods discussed above. Whether τ_{DS} is of interest will necessarily depend on each specific application, but one advantage of focusing on this parameter is that no parametric extrapolation is needed because the object of interest is no longer the fundamentally (nonparametrically) unidentified parameter $\tau_{\text{SRD}} = E[Y_1 | X = c] - E[Y_0 | X = c]$ (for further discussion, illustrations, and references, see Cattaneo et al. 2018b).

While the RD design has recently become one of the leading program evaluation methods, there are still several important methodological challenges when it comes to its implementation and interpretation. To close this section, we briefly mention two of these outstanding research avenues. First, a crucial open question concerns the extrapolation of RD treatment effects. The RD design often offers very credible and robust results for the local subpopulation of units whose scores are near or at the cutoff but is not necessarily informative about the effects at score values far from the cutoff. In other words, while RD treatment effects are regarded as having high internal validity, their external validity is usually suspect. The lack of external validity is, of course, a major cause of concern from a program evaluation and policy prescription perspective. While there is some very recent work that addresses the issue of extrapolation of RD treatment effects (e.g., Angrist & Rokkanen 2015, Bertanha & Imbens 2017, Cattaneo et al. 2017b, Dong & Lewbel 2015), more work is certainly needed, covering both methodological and empirical issues.

A second important open area of research concerns the formalization and methodological analysis of recently developed RD-like research designs. One prominent example is the recent literature on bunching and density discontinuities (Jales & Yu 2017, Kleven 2016), where the objects of interest are related to discontinuities and other sharp changes in a probability density function. Another example is the dynamic RD design (e.g., Cellini et al. 2010), where different units cross the cutoff point at different times, and therefore, other parameters of interest, as

well as identifying assumptions, are considered. These and other RD-like designs have several common features and can be analyzed by employing ideas and tools from the classical RD literature. So far, most of the work in this area has been primarily empirical, but it contains interesting methodological ideas that should now be formalized and analyzed in a principled way. Important issues remain unresolved, ranging from core identification questions to case-specific empirical implementation methods.

8. THE ROAD AHEAD

In this review, we provide an overview of some of the most common econometric methods for program evaluation, covering ideas and tools related to randomized experiments, selection on observables, difference in differences and synthetic controls, IV, and RD designs. We also discuss ongoing developments, as well as some open questions and specific problems, related to these policy evaluation methods.

Our review is, of course, far from exhaustive in terms of both depth within the topics covered and scope for program evaluation methodology. This large literature continues to evolve and adapt to new empirical challenges, and thus, much more methodological work is needed to address the wide range of old and new open problems. We finish by offering a succinct account of some of these problems.

An important recent development that has had a profound impact on the program evaluation literature is the arrival of new data environments (Mullainathan & Spiess 2017). The availability of big data has generated a need for methods able to cope with data sets that are either too large or too complex to be analyzed using standard econometric methods. Of particular importance is the role of administrative records and of large data sets collected by automated systems. These are, in some cases, relatively new sources of information and pose challenges in terms of identification, estimation, and inference. Model selection, shrinkage, and empirical Bayes approaches are particularly useful in this context (e.g., Abadie & Kasy 2017, Efron 2012), although these methods have not yet been fully incorporated into the program evaluation toolkit. Much of the current research in this area develops machine learning methods to estimate heterogeneous treatment effects in contexts with many covariates (see, e.g., Athey & Imbens 2016, Taddy et al. 2016, Wager & Athey 2018).

Also potentiated by the rise of new, big, complex data is the very recent work on networks, spillovers, social interactions, and interference (Graham 2015, Graham & de Paula 2018). While certainly of great importance for policy, these research areas are still evolving and not yet fully incorporated into the program evaluation literature. Bringing developments in these relatively new areas to the analysis and interpretation of policy and treatment effects is important to improve policy prescriptions.

Finally, because of space limitations, some important topics are covered in this review. Among them are mediation analysis (Lok 2016, VanderWeele 2015), dynamic treatment effects and duration models (Abbring & Heckman 2007, Abbring & Van den Berg 2003), bounds and partial identification methods (Manski 2008, Tamer 2010), and optimal design of policies (Hirano & Porter 2009, Kitagawa & Tetenov 2018). These are also important areas of active research in the econometrics of program evaluation.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

We thank Josh Angrist, Max Farrell, Guido Imbens, Michael Jansson, Jérémy L'Hour, Judith Lok, Xinwei Ma, and Rocio Titiunik for thoughtful comments. We gratefully acknowledge financial support from the National Science Foundation grants SES-1459931 to M.D.C. and SES-1756692 to A.A.

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